Daniel Sandi Bratanata

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PROFILE

I'm an undergraduate Information Systems student with strong machine learning and deep learning experience. Currently a Research Assistant that developing a multimodal emotion-detection system using Kolmogorov–Arnold Networks and LLM embeddings (QWEN, BGE-M3) for a Scopus Q2 publication,and also Data Analyst at Ajinomoto Indonesia. Proven technical ability through national competitions, and skilled in end-to-end ML pipelines, from NLP architecture design to deploying models on Hugging Face.

EXPERIENCES

Data Analyst Intern – Ajinomoto Mojokerto Factory **Feb. 2026 – July. 2026**

- Developed a Geospatial Dashboard Application using Django to support sales targeting, including end-to-end data pipeline development and geocoding integration using Bing Maps API for Agridev Department Business Development Section.
- Designed and implemented a PowerBI Dashboard to support avaibility and realtime update data through 21 Key Performance Indicator for 5 Product(AMINA,AJIFOL,GCC-MIX,AJIBIO,FML) monitoring that useful for 4 section (Business Development,Prod L,Prod S,Sales & Marketing)
- Built and managed spatial data processing using PostGIS, enabling efficient storage, querying, and analysis of geospatial data.

AI Engineer Intern – PT.Mandira Abadi **Jan. 2026 – Feb. 2026**

- Developed an embedding-based similarity pipeline using BGE-M3 to identify matching hotels across multiple data sources, enabling accurate entity resolution and supporting competitive pricing analysis.
- Implemented feature fusion combining TF-IDF and BGE-M3 embeddings, enhanced with similarity metrics including Jaccard, Dice, and Cosine similarity for robust matching performance.
- Achieved 99% precision in hotel similarity detection under the supervision of Benedictus Kent Rachmat (Mandira Tech Advisor and AI Researcher at Schlumberger Paris).

AI Research Assistant - Institut Teknologi Sepuluh Nopember **Aug. 2025 – May. 2026**

- Conducted web-scraping using Selenium to performed preliminary research proposing the use of Kolmogorov–Arnold Network (KAN) and the CBLMP mechanism for emotion classification.
- Enhancing domain dataset with change and combine data with other data with same label and get best lambda from domain data and external data to generate new data with larger amount using SMOTExT that utilize xRAG(Mistral 7B)
- Developed a multimodal model combining text embeddings (Qwen-0.6B, BGE M3, Jina V3) and image embeddings(DINO-V3) for early detection of adolescent mental-health issues; research targeted for publication in the International Journal of Intelligent and Engineering Systems (Scopus Q2).

Programming Teacher - Timedoor **Apr. 2025 – Dec. 2025**

- Taught Web Programming (HTML, CSS, JavaScript) to 6+ classes, providing hands-on, industry-relevant training.
- Delivered Artificial Intelligence Fundamentals, including practical image-processing projects such as eye and lip detection using modern AI techniques.

EDUCATION

2023 - Present

Institut Teknologi Sepuluh Nopember
Bachelor of Information System

CGPA– 3.58
Transcript

ORGANIZATIONS

Data Scientist Staff - AVALON AI (PT.Automasi Visi Analitika) **Nov. 2024 – Nov. 2025**

- Developed an XGBoost-based machine learning model to predict fraudulent transactions using features such as time, location, and store data, improving detection accuracy.
- Built a CNN-based multiclass image classification model to identify hoodies and t-shirts in 5 colors, enabling efficient automated product recognition.

Senior Backend Engineer – Information System Expo **Apr. 2025 – Dec. 2025**

- Designed database entity models and developed key APIs using the Gonic framework in Golang.
- Architected and optimized a PostgreSQL infrastructure that reliably supported three consecutive high-traffic events, scaling seamlessly to serve 150+ concurrent participants with zero downtime and consistent sub-second query performance.

SKILLS & TOOLS

Hard Skills

- Programming & Data: Python, SQL,Fast API
- Machine Learning & Deep Learning: TensorFlow, Scikit-learn, PyTorch,Sentence Transformer,OpenCV,FAISS(Vector Similarity)
- Data Visualization: PowerBI, Matplotlib, Seaborn, Plotly
- Data Engineering: PostgreSQL, Pandas, NumPy
- Tools: Anaconda, Visual Studio Code

Soft Skills

- Communication
- Teamwork
- Problem-Solving
- Adaptability
- Time Management

Languages

- Indonesia (Native)
- English (Proficient)

ACHIEVEMENTS

3rd Place Data Competition - Multimedia Nusantara University **Oct. 2025**

- Analyzed 2020 Open Source Transjakarta dataset to extract insights and address key issues in predicting hourly passengers per bus stop.
- Secured a top-tier competitive ranking (3rd place) by engineering a high-precision XGBoost model that delivered industry-leading predictive accuracy, setting a new performance benchmark.

2nd Place Machine Learning Competition - Airlangga University **Sept. 2025**

- Achieved 2nd place out of 200 teams in a Machine Learning Competition at Airlangga University by analyzing prison verdict documents using NLP.
- Engineered a sentence complexity classifier to streamline content processing, enabling 20% faster document handling and more efficient resource allocation for high-volume text analytics.

6th Place Data Mining Competition –Telkom University **Nov. 2025**

- Achieved 6th place out of 103 teams by developing an end-to-end data analysis pipeline for undersea mammal classification.
- Engineered the WhaleNet computer vision system to solve the manual bottleneck in marine image analysis, achieving over 95% identification accuracy and reducing data-to-insight timelines by 40%.

7th Place Data Analyst FIND-IT Competition – Gadjah Mada University **Apr. 2025**

- Ranked among the top 10 finalists out of 1,181 teams by developing end-to-end data analysis pipelines and predictive classification models.
- Engineered a predictive model using tabular app metadata to classify COPPA compliance risk, transforming manual regulatory review into an automated, scalable audit system that mitigates legal exposure.

Certifications

- Supervised Machine Learning and Unsupervised Machine Learning, DeepLearning.AI And Stanford (92.46)
- Linear Algebra for Machine Learning and Data Science, DeepLearning.AI And Stanford (93.32)
- Introduction to SAP S/4HANA with GBI 4.2, Department of Information Systems Institut Teknologi Sepuluh Nopember (84.83)